

Structural Dynamics

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Title: Forced Vibration Systems (SDOF)

Question: An undamped system has, $m = 10$ kg, $k = 1000$ N/m. If the system is subjected to initial displacement and velocity of 0.1 m and 10 m/s respectively, and it is excited by a sinusoidal harmonic force having an amplitude $F_0 = 50$ N.

Answer:

→ Natural Frequency: $\omega_n = \sqrt{\frac{k}{m}}$

→ Frequency Ratio: $\beta = r = \frac{\omega}{\omega_n}$

Equation of Motion

$$\rightarrow m\ddot{x} + kx = F_0 \sin(\omega t)$$

→ Static Deflection: $\delta_{st} = \frac{F_0}{k}$

→ Amplitude: $X = \frac{F_0/k}{|1 - \beta^2|} = \frac{\delta_{st}}{|1 - \beta^2|}$

→ Magnification Factor: $R_d = \frac{1}{|1 - \beta^2|}$

→ Steady – State Displacement: $x_{ss}(t) = X \sin(\omega t - \phi)$

Transient Response (Homogeneous Solution)

$$x_h(t) = A \cos(\omega_n t) + B \sin(\omega_n t)$$

Initial Condition Coefficients:

$$A = x_0 - X \sin(-\phi)$$

$$B = \frac{v_0 - X\omega \cos(-\phi)}{\omega_n}$$

Total Response:

$$\rightarrow x(t) = x_h(t) + x_{ss}(t)$$

$$\rightarrow x(t) = A \cos(\omega_n t) + B \sin(\omega_n t) + X \sin(\omega t - \phi)$$

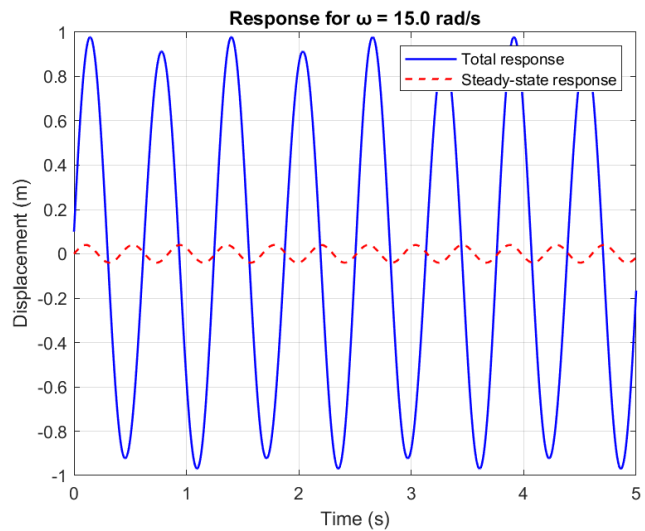
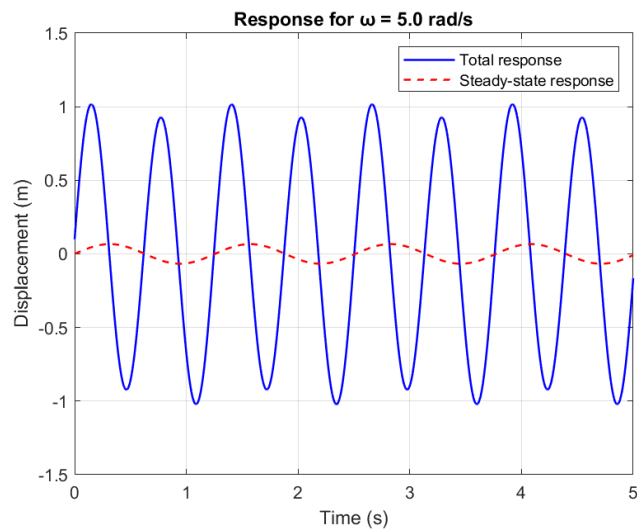
Results:

For $\omega = 5.0$ rad/s:

- $\beta = 0.500$
- $X = 0.0667$ m
- $R_d = 1.333$
- $\phi = 0.000$ rad

For $\omega = 15.0$ rad/s:

- $\beta = 1.500$
- $X = 0.0400$ m
- $R_d = 0.800$
- $\phi = 3.142$ rad



Question: If the same system given in Question 1 has a damping of $c = 80 \text{ Ns/m}$, (a) Plot the total and steady-state damped response for excitation frequencies of $\omega = 5 \text{ rad/s}$ and 15 rad/s . (b) Determine (i) the amplitude of the forced response, (ii) the magnification factor R_d and (iii) phase angle ϕ the damped system for both the forcing frequencies.

Answer:

$$\rightarrow \text{Natural Frequency: } \omega_n = \sqrt{\frac{k}{m}}$$

$$\rightarrow \text{Critical Damping Coefficient: } c_c = 2\sqrt{km} = 2m\omega_n$$

$$\rightarrow \text{Damping Ratio: } \zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}}$$

$$\rightarrow \text{Damped Natural Frequency: } \omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (\text{for } \zeta < 1)$$

$$\rightarrow \text{Frequency Ratio: } \beta = r = \frac{\omega}{\omega_n}$$

Equation of Motion

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega t)$$

$$\rightarrow \text{Static Deflection: } \delta_{st} = \frac{F_0}{k}$$

$$\rightarrow \text{Amplitude: } X = \frac{F_0/k}{\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}} = \frac{\delta_{st}}{\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}}$$

$$\rightarrow \text{Phase Angle: } \phi = \arctan\left(\frac{2\zeta\beta}{1 - \beta^2}\right) = \tan^{-1}\left(\frac{2\zeta\beta}{1 - \beta^2}\right)$$

$$\rightarrow \text{Magnification Factor: } R_d = \frac{X}{\delta_{st}} = \frac{1}{\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}}$$

Steady-State Displacement:

$$\rightarrow x_{ss}(t) = X \sin(\omega t - \phi)$$

Transient Response (Underdamped: $\zeta < 1$)

$$\rightarrow x_h(t) = e^{-\zeta\omega_n t} [A \cos(\omega_d t) + B \sin(\omega_d t)]$$

Initial Condition Coefficients:

$$A = x_0 - X \sin(-\phi)$$

$$B = \frac{v_0 + \zeta\omega_n A - X\omega \cos(-\phi)}{\omega_d}$$

Total Response (Underdamped)

$$\rightarrow x(t) = x_h(t) + x_{ss}(t)$$

$$\rightarrow x(t) = e^{-\zeta\omega_n t} [A \cos(\omega_d t) + B \sin(\omega_d t)] + X \sin(\omega t - \phi)$$

Transient Response (Critically Damped: $\zeta = 1$)

$$\rightarrow x_h(t) = e^{-\omega_n t} (C_1 + C_2 t)$$

Transient Response (Overdamped: $\zeta > 1$)

$$\rightarrow x_h(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

where: $\lambda_1 = -\zeta\omega_n + \omega_n \sqrt{\zeta^2 - 1}$

$$\rightarrow \lambda_2 = -\zeta\omega_n - \omega_n \sqrt{\zeta^2 - 1}$$

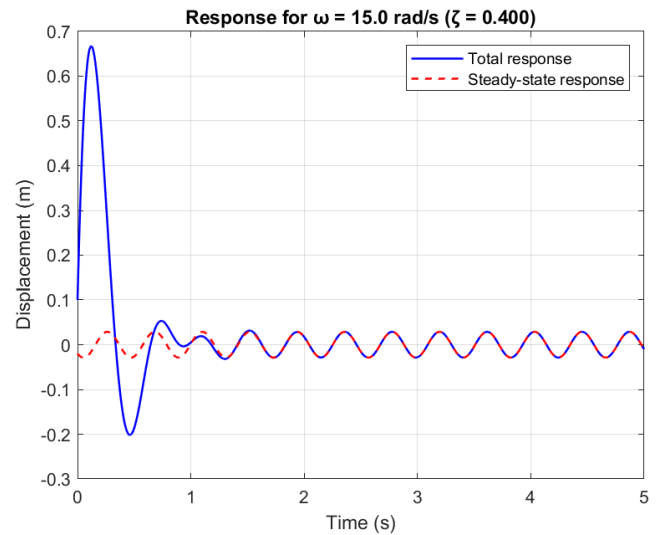
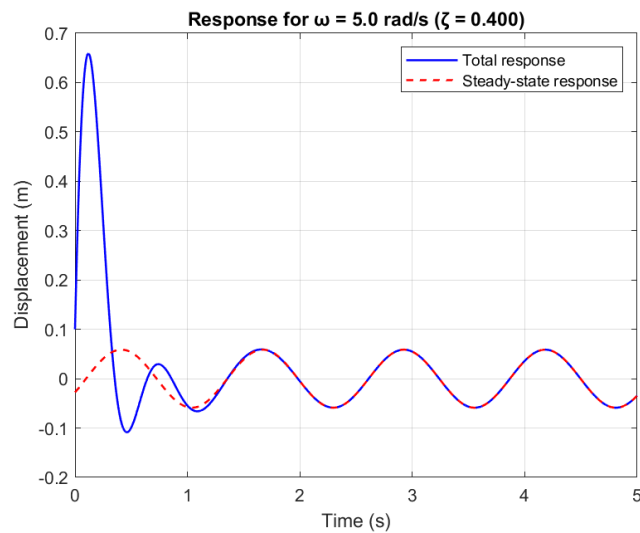
Results:

For $\omega = 5.0$ rad/s:

- $\beta = 0.5000$
- $X = 0.058824$ m
- $Rd = 1.1765$
- $\phi = 0.4900$ rad (28.07°)

For $\omega = 15.0$ rad/s:

- $\beta = 1.5000$
- $X = 0.028855$ m
- $R_d = 0.5771$
- $\varphi = 2.3766$ rad (136.17°)



MATLAB Code

```
clc; clear; close all;

% Given data
m = 10;           % kg
k = 1000;         % N/m
F0 = 50;          % N
x0 = 0.1;         % m
v0 = 10;          % m/s
wn = sqrt(k/m);   % Natural frequency (rad/s)

% Excitation frequencies
omega_list = [5 15];

for i = 1:length(omega_list)
    omega = omega_list(i);

    % Frequency ratio
    beta = omega / wn;

    % Steady-state amplitude (undamped)
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X = (F0/k) / (1 - beta^2);

% Phase angle
if omega < wn
    phi = 0;
else
    phi = pi;
end

% Magnification factor
Rd = 1 / abs(1 - beta^2);

% Time vector
t = linspace(0, 5, 1000);

% Homogeneous solution coefficients from initial conditions
A = x0 - X * sin(-phi);
B = (v0 - X * omega * cos(-phi)) / wn;

% Total and steady-state responses
x_total = A * cos(wn * t) + B * sin(wn * t) + X * sin(omega * t - phi);
x_steady = X * sin(omega * t - phi);

% Display results
fprintf('\nFor  $\omega = %.1f$  rad/s:\n', omega);
fprintf('  $\beta = %.3f$ \n', beta);
fprintf(' X = %.4f m\n', abs(X));
fprintf(' Rd = %.3f\n', Rd);
fprintf('  $\phi = %.3f$  rad\n', phi);

% Plotting
figure;
plot(t, x_total, 'b', 'LineWidth', 1.2); hold on;
plot(t, x_steady, 'r--', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Displacement (m)');
title(sprintf('Response for  $\omega = %.1f$  rad/s', omega));
legend('Total response', 'Steady-state response');

end

% Given data
m = 10; % kg
k = 1000; % N/m
c = 80; % N·s/m (damping coefficient)
F0 = 50; % N
x0 = 0.1; % m
v0 = 10; % m/s
wn = sqrt(k/m); % Natural frequency (rad/s)

% Damping parameters
zeta = c / (2 * sqrt(k * m)); % Damping ratio
wd = wn * sqrt(1 - zeta^2); % Damped natural frequency

% Excitation frequencies

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omega_list = [5 15];

for i = 1:length(omega_list)
    omega = omega_list(i);

    % Frequency ratio
    beta = omega / wn;

    % Steady-state amplitude (damped)
    X = (F0/k) / sqrt((1 - beta^2)^2 + (2*zeta*beta)^2);

    % Phase angle
    phi = atan2(2*zeta*beta, 1 - beta^2);

    % Magnification factor
    Rd = 1 / sqrt((1 - beta^2)^2 + (2*zeta*beta)^2);

    % Time vector
    t = linspace(0, 5, 1000);

    % Homogeneous solution coefficients from initial conditions
    A = x0 - X * sin(-phi);
    B = (v0 + zeta*wn*A - X*omega*cos(-phi)) / wd;

    % Total and steady-state responses
    x_transient = exp(-zeta*wn*t) .* (A * cos(wd*t) + B * sin(wd*t));
    x_steady = X * sin(omega * t - phi);
    x_total = x_transient + x_steady;

    % Display results
    fprintf('\nFor  $\omega = %.1f$  rad/s:\n', omega);
    fprintf('   $\beta = %.4f$ \n', beta);
    fprintf('   $X = %.6f$  m\n', X);
    fprintf('   $R_d = %.4f$ \n', Rd);
    fprintf('   $\phi = %.4f$  rad ( $%.2f^\circ$ )\n', phi, rad2deg(phi));

    % Plotting
    figure;
    plot(t, x_total, 'b', 'LineWidth', 1.2); hold on;
    plot(t, x_steady, 'r--', 'LineWidth', 1.2);
    grid on;
    xlabel('Time (s)');
    ylabel('Displacement (m)');
    title(sprintf('Response for  $\omega = %.1f$  rad/s ( $\zeta = %.3f$ )', omega, zeta));
    legend('Total response', 'Steady-state response');
end

```